Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

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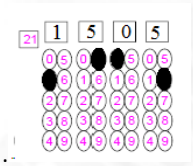
SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 09-01-2026**

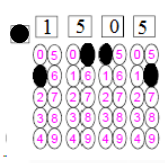
Time: 09.00Am to 12.00Pm **GTM-11** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below





Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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09-01-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main_GTM-11_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let $a_1, a_2, a_3, \ldots\ldots$ be a G.P. of increasing positive numbers. Let the sum of its 6th and 8th terms be 2 and the product of its 3rd and 5th terms be 1/9. Then $6(a_2 + a_4)(a_4 + a_6)$ is equal to

1) $2\sqrt{2}$ **2)** 2 **3)** $3\sqrt{3}$ **4)** 3

2. The random variable X follows binomial distribution B (n, p), for which the difference of the mean and the variance is 1. If $2P(X=2)=3P(X=1)$, *then* $n^2P(X>1)$ is equal to

1) 12 **2)** 15 **3)** 11 **4)** 16

3. Let $S=\left\{z\in C:\bar{z}=i\left(z^2+\mathrm{Re}(\bar{z})\right)\right\}$. Then $\sum_{z\in S}|z|^2$ is equal to

1) 5/2 **2)** 4 **3)** 3 **4)** 7/2

4. Let $|\vec{a}|=2, |\vec{b}|=3$ and the angle between the vectors $\vec{a}$ and $\vec{b}$ be $\frac{\pi}{4}$. Then $|(\vec{a}+2\vec{b})\times(2\vec{a}-3\vec{b})|^2$ is equal to

1) 441 **2)** 882 **3)** 841 **4)** 482

5. The value of $\dfrac{e^{-\pi/4}+\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}{\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49}x+\tan^{51}x\right)dx}$ is

1) 51 **2)** 25 **3)** 50 **4)** 49

6. If $\lim\limits_{x\to 0}\dfrac{e^{ax}-\cos(bx)-\frac{cxe^{-cx}}{2}}{1-\cos(2x)}=17$, then $5a^2+b^2$ is equal to

1) 64 **2)** 76 **3)** 68 **4)** 72

7. Let α, β be the roots of the equation $x^2-\sqrt{2}x+2=0$. Then $\alpha^{14}+\beta^{14}$ is equal to

1) -64 **2)** $-64\sqrt{2}$ **3)** $-128\sqrt{2}$ **4)** -128

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8. The line, that is coplanar to the line $\frac{x+3}{-3}=\frac{y-1}{1}=\frac{z-5}{5}$, is

1) $\frac{x-1}{-1}=\frac{y-2}{2}=\frac{z-5}{5}$ **2)** $\frac{x+1}{1}=\frac{y-2}{2}=\frac{z-5}{5}$

3) $\frac{x+1}{-1}=\frac{y-2}{2}=\frac{z-5}{4}$ **4)** $\frac{x+1}{-1}=\frac{y-2}{2}=\frac{z-5}{5}$

9. All words, with or without meaning, are made using all the letters of the word MONDAY. These words are written as in a dictionary with serial numbers. The serial number of the word MONDAY is

1) 327 **2)** 326 **3)** 328 **4)** 324

10. The coefficient of x^5 in the expansion of $\left(2x^3-\frac{1}{3x^2}\right)^5$ is

1) 26/3 **2)** 9 **3)** 80/9 **4)** 8

11. Let for $A=\begin{bmatrix}1 & 2 & 3\\ \alpha & 3 & 1\\ 1 & 1 & 2\end{bmatrix}, |A|=2$. If $|2\text{adj}(2\text{adj}(2A))|=32^n$, then $3n+\alpha$ is equal to

1) 10 **2)** 12 **3)** 9 **4)** 11

12. If the system of equations $2x+y-z=5$, $2x-5y+\lambda z=\mu$, $x+2y-5z=7$ has infinitely many solutions, then $(\lambda+\mu)^2+(\lambda-\mu)^2$ is equal to

1) 912 **2)** 920 **3)** 916 **4)** 904

13. Let the centre of a circle C be (α,β) and its radius $r < 8$. Let $3x + 4y = 24$ and $3x - 4y = 32$ be two tangents and $4x + 3y = 1$ be a normal to C. Then $(\alpha-\beta+r)$ is equal to

1) 9 **2)** 7 **3)** 6 **4)** 5

14. The area of the region $\left\{(x,y): x^2 \le y \le \left|x^2-4\right|, y \ge 1\right\}$ is

1) $\frac{4}{3}\left(4\sqrt{2}+1\right)$ **2)** $\frac{4}{3}\left(4\sqrt{2}-1\right)$ **3)** $\frac{3}{4}\left(4\sqrt{2}+1\right)$ **4)** $\frac{3}{4}\left(4\sqrt{2}-1\right)$

15. The plane, passing through the points (0,-1,2) and (-1,2,1) and parallel to the line passing through (5,1,-7) and (1,-1,-1), also passes through the point

1) (-2, 5, 0) **2)** (1, -2, 1) **3)** (2, 0, 1) **4)** (0, 5, -2)

16. Let (α,β) be the centroid of the triangle formed by the lines $15x-y=82, 6x-5y=-4$ and $9x+4y=17$. Then $\alpha+2\beta$ and $2\alpha-\beta$ are the roots of the equation

1) $x^2 - 10x + 25 = 0$ **2)** $x^2 - 14x + 48 = 0$

3) $x^2 - 13x + 42 = 0$ **4)** $x^2 - 7x + 12 = 0$

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17. Let N be the foot of perpendicular from the point P(1, -2, 3) on the line passing through the points (4, 5, 8) and (1, -7, 5). Then the distance of N from the plane 2x-2y+z+5=0 is

1) 6 **2)** 8 **3)** 7 **4)** 9

18. The range of $f(x)=4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is

1) $[0, 2\pi]$ **2)** $[0, \pi]$ **3)** $[0, \pi)$ **4)** $[0, 2\pi)$

19. Let for a triangle ABC,

$\overrightarrow{AB}=-2\hat{i}+\hat{j}+3\hat{k}$

$\overrightarrow{CB}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$

$\overrightarrow{CA}=4\hat{i}+3\hat{j}+\delta\hat{k}$

If $\delta>0$ and the area of the triangle ABC is $5\sqrt{6}$, then $\overrightarrow{CB}\cdot\overrightarrow{CA}$ is equal to

1) 108 **2)** 120 **3)** 54 **4)** 60

20. If the shortest distance of the parabola $y^2=4x$ from the centre of the circle $x^2+y^2-4x-16y+64=0$ is d, then d^2 is equal to :

1) 16 **2)** 20 **3)** 36 **4)** 24

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. The foci of a hyperbola are (±2, 0) and its eccentricity is 3/2. A tangent, perpendicular to the line 2x + 3y = 6, is drawn at a point in the first quadrant on the hyperbola. If the intercepts made by the tangent on the x- and y-axes are a and b respectively, then |6a|+|5b| is equal to_____

22. Let $f_n=\int_0^{\frac{\pi}{2}}\left(\sum_{k=1}^{n}\sin^{k-1}x\right)\left(\sum_{k=1}^{n}(2k-1)\sin^{k-1}x\right)\cos x\,dx, n\in N$. Then $f_{21}-f_{20}$ is equal to

23. Let [α] denote the greatest integer ≤α. Then $\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+.....+\left[\sqrt{120}\right]$ is equal to ___

24. If y = y(x) is the solution of the differential equation $\frac{dy}{dx}+\frac{4x}{(x^2-1)}y=\frac{x+2}{(x^2-1)^{5/2}}, x>1$ such that $y(2)=\frac{2}{9}\log_e(2+\sqrt{3})$ and $y(\sqrt{2})=\alpha\log_e(\sqrt{\alpha}+\beta)+\beta-\sqrt{\gamma}, \alpha,\beta,\gamma\in N$, then αβγ is equal to ____

25. Let A = {-4, -3, -2, 0, 1, 3, 4} and $R=\{(a,b)\in A\times A: b=|a| or\ b^2=a+1\}$ be a relation on A. Then the minimum number of elements, that must be added to the relation R so that it becomes reflexive and symmetric, is _____

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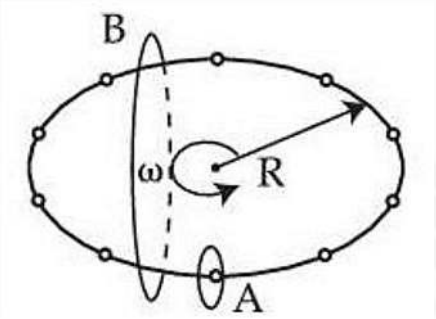
PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26.



N equally spaced charges each of value q, are placed on a circle of radius R. The circle rotates about its axis with an angular velocity ω as shown in the figure. A bigger Amperian loop B encloses the whole circle where as a smaller Amperian loop A encloses a small segment. The difference between enclosed currents, $I_A - I_B$, for the given Amperian loops is

1) $\frac{N^2}{2\pi}q\omega$ **2)** $\frac{2\pi}{N}q\omega$ **3)** $\frac{N}{2\pi}q\omega$ **4)** $\frac{N}{\pi}q\omega$

27. In photoelectric effect, the stopping potential $(V_0)v/s$ frequency (v) curve is plotted. (h is the Planck's constant and ϕ_0 is work function of metal)

(A) V_0 v/s v is linear

(B) The slope of V_0 v/s v curve $= \frac{\phi_0}{h}$

(C) h constant is related to the slope of V_0 v/s v line

(D) The value of electric charge of electron is not required to determine h using the V_0 v/s v curve.

(E) The work function can be estimated without knowing the value of h.

Choose the correct answer from the options given below.

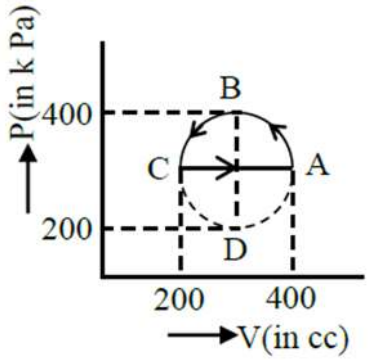
1) (A),(B) and (C) only **2)** (C) and (D) only

3) (A),(C) and (E) only **4)** (D) and (E) only

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28. The magnitude of heat exchanged by a system for the given cyclic process ABCA (as shown in figure) is (in SI unit)

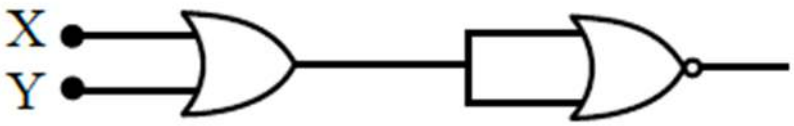


1) 10π **2)** 5π **3)** zero **4)** 40π

29. A photograph of a landscape is captured by a drone camera at a height of 18 km. The size of the camera film is 2 cm × 2 cm and the area of the landscape photographed is 400 km^2. The focal length of the lens in the drone camera is :

1) 1.8 cm **2)** 2.8 cm **3)** 2.5 cm **4)** 0.9 cm

30. The output of the circuit is low (zero) for :



(A) X = 0, Y= 0 (B) X = 0, Y= 1

(C) X = 1, Y= 0 (D) X = 1, Y= 1

Choose the correct answer from the options given below :

1) (A), (C) and (D) only **2)** (A), (B) and (C) only

3) (B), (C) and (D) only **4)** (A), (B) and (D) only

31. The temperature of a body in air falls from 40°C to 24°C in 4 minutes. The temperature of the air is 16°C. The temperature of the body in the next 4 minutes will be :

1) $\frac{14}{3}$ °C **2)** $\frac{28}{3}$ °C **3)** $\frac{56}{3}$ °C **4)** $\frac{42}{3}$ °C

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32. The energy E and momentum p of a moving body of mass m are related by some equation. Given that c represents the speed of light, identify the correct equation.

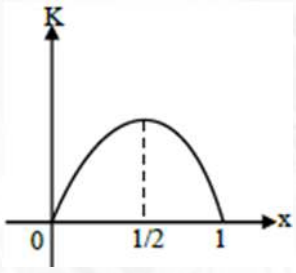
1) $E^2 = pc^2 + m^2c^4$ **2)** $E^2 = pc^2 + m^2c^2$

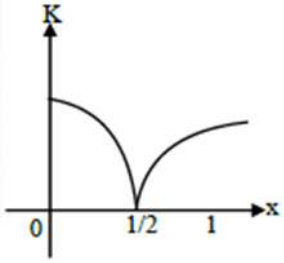
3) $E^2 = p^2c^2 + m^2c^2$ **4)** $E^2 = p^2c^2 + m^2c^4$

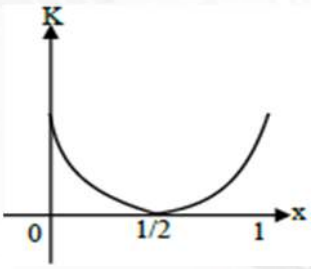
33. A small uncharged conducting sphere is placed in contact with an identical sphere but having 4×10^{-8} C charge and then removed to a distance such that the force of repulsion between them is 9×10^{-3}N. The distance between them is (Take $\frac{1}{4\pi\varepsilon_0}$ as 9×10^9 in SI units)

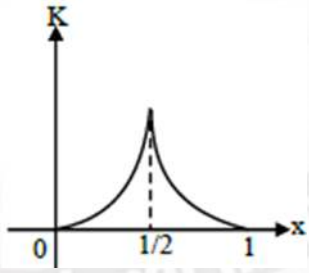
1) 2 cm **2)** 3 cm **3)** 4 cm **4)** 1 cm

34. A particle oscillates along the x-axis according to the law, $x(t) = x_0 \sin^2\left(\frac{t}{2}\right)$ where x_0 = 1 m. The kinetic energy (K) of the particle as a function of x is correctly represented by the graph.

1)


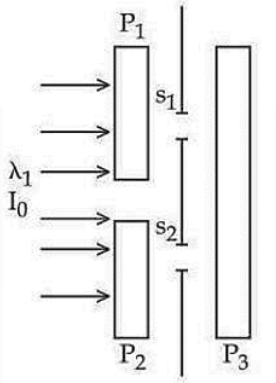
2)


3)


4)


35. In a Young's double slit experiment, three polarizers are kept as shown in the figure. The transmission axes of P_1 and P_2 are orthogonal to each other. The polarizer P_3 covers both the slits with its transmission axis at 45° to those of P_1 and P_2. An unpolarized light of wavelength λ and intensity I_0 is incident on P_1 and P_2. The intensity at a point after P_3 where the path difference between the light waves from s_1 and s_2 is $\frac{\lambda}{3}$, is

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1) $\frac{I_0}{2}$ **2)** $\frac{I_0}{4}$ **3)** I_0 **4)** $\frac{I_0}{3}$

36. Young's double slit interference apparatus is immersed in a liquid of refractive index 1.44. It has slit separation of 1.5mm. The slits are illuminated by a parallel beam of light whose wavelength in air is 690 nm. The fringe-width on a screen placed behind the plane of slits at a distance of 0.72m, will be :

1) 0.23 mm **2)** 0.33 mm **3)** 0.63 mm **4)** 0.46 mm

37. Arrange the following in the ascending order of wavelength (λ):

(A) Microwaves (λ_1)

(B) Ultraviolet rays (λ_2)

(C) Infrared rays (λ_3)

(D) X-rays (λ_4)

Choose the most appropriate answer from the options given below :-

1) $\lambda_4 < \lambda_3 < \lambda_2 < \lambda_1$ **2)** $\lambda_3 < \lambda_4 < \lambda_2 < \lambda_1$ **3)** $\lambda_4 < \lambda_2 < \lambda_3 < \lambda_1$ **4)** $\lambda_4 < \lambda_3 < \lambda_1 < \lambda_2$

38. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A) : A electron in a certain region of uniform magnetic field is moving with constant velocity in a straight line path.

Reason (R) : The magnetic field in that region is along the direction of velocity of the electron.

In the light of the above statements, choose the correct answer from the options given below:

1) (A) is false but (R) is true

2) Both (A) and (R) are true and (R) is the correct explanation of (A)

3) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

4) (A) is true but (R) is false

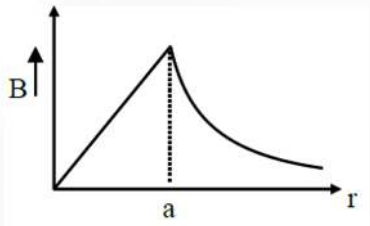
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39. A solid sphere is rolling without slipping on a horizontal plane. The ratio of the linear kinetic energy of the centre of mass of the sphere and rotational kinetic energy is :

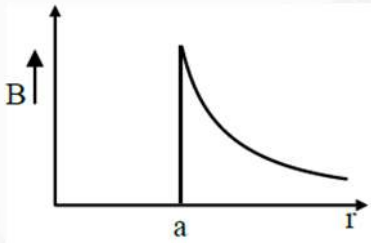
1) 2/5 **2)** 5/2 **3)** 3/4 **4)** 4/3

40. A long straight wire of a circular cross-section with radius 'a' carries a steady current I. The current I is a uniformly distributed across this cross-section. The plot of magnitude of magnetic field B with distance r from the centre of the wire is given by

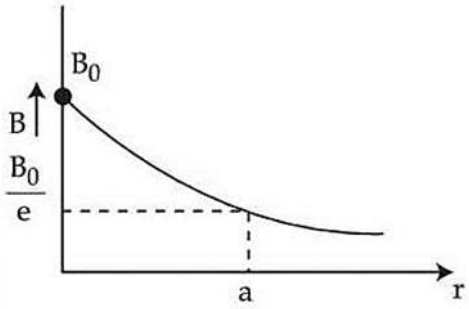
1)



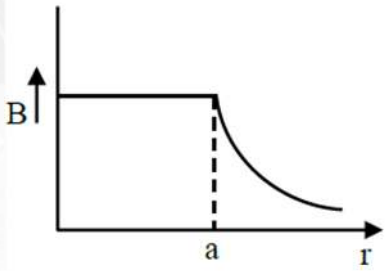
2)



3)



4)



41. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A) : In an insulated container, a gas is adiabatically shrunk to half of its initial volume. The temperature of the gas decreases.

Reason (R) : Free expansion of an ideal gas is an irreversible and an adiabatic process.

In the light of the above statement, choose the correct answer from the options given below :

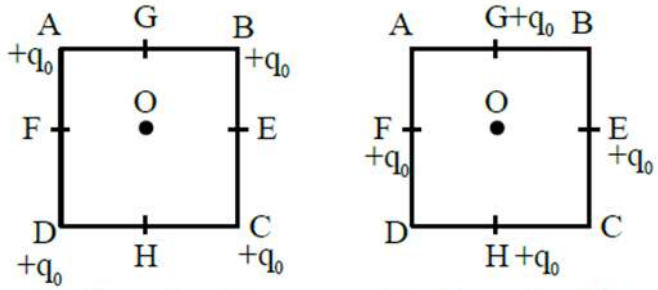
1) Both (A) and (R) are true and (R) is the correct explanation of (A)

2) (A) is true but (R) is false

3) (A) is false but (R) is true

4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

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42. In the first configuration (1) as shown in the figure, four identical charges (q_0) are kept at the corners A, B, C and D of square of side length 'a'. In the second configuration (2), the same charges are shifted to mid points G,E,H and F, of the square. If $K = \frac{1}{4\pi\varepsilon_0}$, the difference between the potential energies of configuration (2) and (1) is given by :

1) $\frac{Kq_0^2}{a}\left(4\sqrt{2}-2\right)$ **2)** $\frac{Kq_0^2}{a}\left(3-\sqrt{2}\right)$ **3)** $\frac{Kq_0^2}{a}\left(4-2\sqrt{2}\right)$ **4)** $\frac{Kq_0^2}{a}\left(3\sqrt{2}-2\right)$

43. The position vector of a moving body at any instant of time is given as $\vec{r} = \left(5t^2\hat{i} - 5t\,\hat{j}\right)$m.
The magnitude and direction of velocity at t=2s is,

1) $5\sqrt{15}$m/s, making an angle of $\tan^{-1}4$ with –ve Y axis

2) $5\sqrt{15}$m/s, making an angle of $\tan^{-1}4$ with +ve X axis

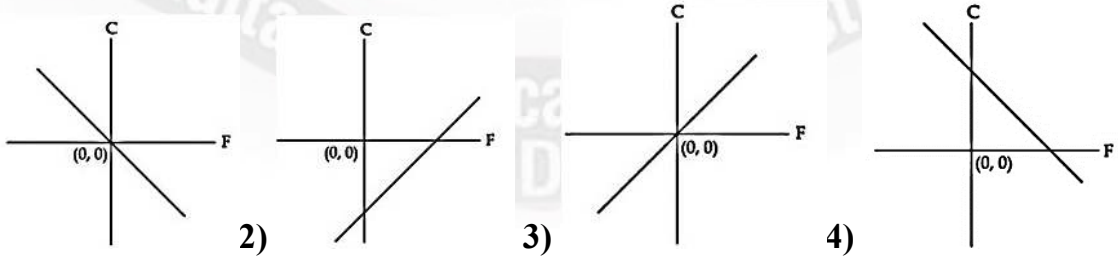
3) $5\sqrt{17}$m/s, making an angle of $\tan^{-1}4$ with –ve Y axis

4) $5\sqrt{17}$m/s, making an angle of $\tan^{-1}4$ with +ve X axis

44. A solid sphere and a hollow sphere of the same mass and of same radius are rolled on an inclined plane. Let the time taken to reach the bottom by the solid sphere and the hollow sphere be t_1 and t_2, respectively, then

1) $t_1 < t_2$ **2)** $t_1 = t_2$ **3)** $t_1 = 2t_2$ **4)** $t_1 > t_2$

45. Which of the following figure represents the relation between Celsius and Fahrenheit temperatures?



1) **2)** **3)** **4)**

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SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

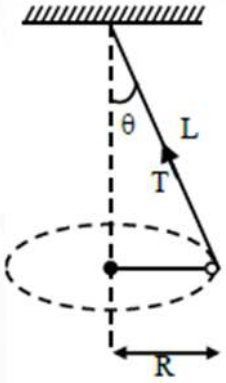
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. The increase in pressure required to decrease the volume of a water sample by 0.2% is $P\times10^{5}Nm^{-2}$. Bulk modulus of water is $2.15\times10^{9}Nm^{-2}$. The value of P is _____.

47. Acceleration due to gravity on the surface of earth is 'g'. If the diameter of earth is reduced to one third of its original value and mass remains unchanged, then the acceleration due to gravity on the surface of the earth is ____g.

48. A tightly wound long solenoid carries a current of 1.5 A. An electron is executing uniform circular motion inside the solenoid with a time period of 75ns. The number of turns per metre in the solenoid is_____.
[Take mass of electron $m_e = 9\times10^{-31}$kg, charge of electron $|q_e| = 1.6\times10^{-19}$C, $\mu_0 = 4\pi\times10^{-7}\frac{N}{A^2}$, $1ns = 10^{-9}s$]]

49.



A string of length L is fixed at one end and carries a mass of M at the other end. The mass makes $\left(\frac{3}{\pi}\right)$ rotations per second about the vertical axis passing through end of the string as shown. The tension in the string is _____ ML.

50. The ratio of the power of a light source S_1 to that the light source S_2 is 2. S_1 is emitting 2×10^{15} photons per second at 600 nm. If the wavelength of the source S_2 is 300 nm, then the number of photons per second emitted by S_2 is_____$\times10^{14}$.

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CHEMISTRY

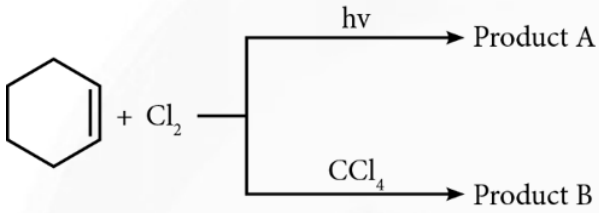
Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

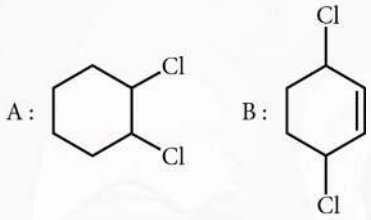
This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

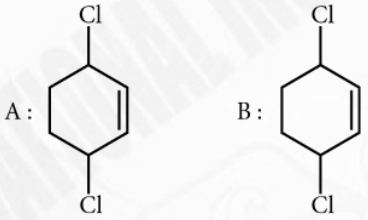
51. Identify product A and product B :



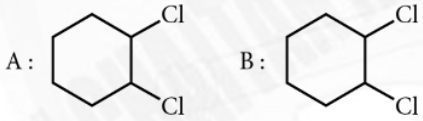
1)



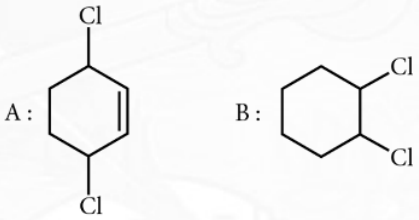
2)



3)



4)



52. Identify the incorrect pair from the following :

1) Fluorspar- BF_3 **2)** Cryolite $-Na_3AlF_6$

3) Fluoroapatite$-3Ca_3(PO_4)_2 \cdot CaF_2$ **4)** Carnallite$-KCl \cdot MgCl_2 \cdot 6H_2O$

53. The interaction between π bond and lone pair of electrons present on an adjacent atom is responsible for

1) Hyperconjugation **2)** Inductive effect

3) Electromeric effect **4)** Resonance effect

54. $KMnO_4$ decomposes on heating at 513K to form O_2 along with

1) MnO_2 & K_2O_2 **2)** K_2MnO_4 & Mn

3) Mn & KO_2 **4)** K_2MnO_4 & MnO_2

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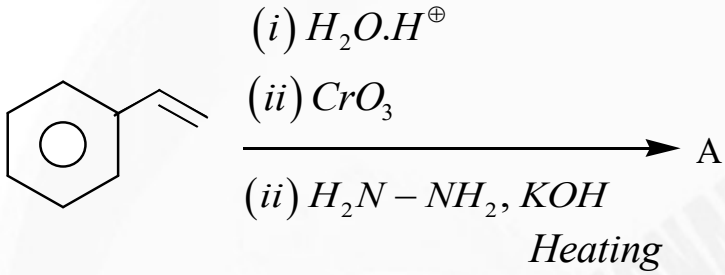
55. In which one of the following metal carbonyls, CO forms a bridge between metal atoms?

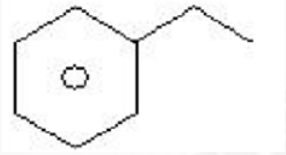
1) $[Co_2(CO)_8]$ **2)** $[Mn_2(CO)_{10}]$ **3)** $[Os_3(CO)_{12}]$ **4)** $[Ru_3(CO)_{12}]$

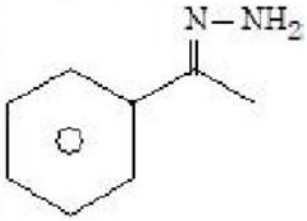
56. Type of amino acids obtained by hydrolysis of proteins is :

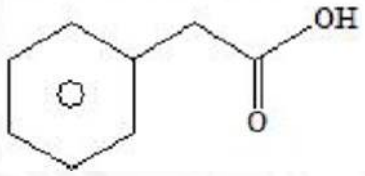
1) β **2)** α **3)** δ **4)** γ

57. The final product A formed in the following multistep reaction sequence is



1) 

2) 

3) 

4) 

58. Which of the following is not correct?

1) ΔG is negative for a spontaneous reaction

2) ΔG is positive for a spontaneous reaction

3) ΔG is zero for a reversible reaction

4) ΔG is positive for a non-spontaneous reaction

59. Chlorine undergoes disproportionation in alkaline medium as shown below:

$$aCl_2(g) + bOH^-(aq) \rightarrow cClO^-(aq) + dCl^-(aq) + eH_2O(l)$$

The values of a, b, c and d in a balanced redox reaction are respectively:

1) 1, 2, 1 and 1 **2)** 2, 2, 1 and 3 **3)** 3, 4, 4 and 2 **4)** 2, 4, 1 and 3

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60. In alkaline medium. MnO_4^- oxidises I^- to

1) IO_4^- **2)** IO^- **3)** I_2 **4)** IO_3^-

61. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The first ionisation enthalpy decreases across a period.

Reason R: The increasing nuclear charge outweighs the shielding across the period.

In the light of the above statements, choose the most appropriate from the options given below:

1) Both A and R are true and R is the correct explanation of A

2) A is true but R is false

3) A is false but R is true

4) Both A and R are true but R is NOT the correct explanation of A

62. Match List I with List II.

	LIST-I (Substances)		**LIST-II (Element Present)**
A	Ziegler catalyst	I	Rhodium
B	Blood Pigment	II	Cobalt
C	Wilkinson catalyst	III	Iron
D	Vitamin B_{12}	IV	Titanium

Choose the correct answer from the options given below:

1) A-II, B-IV, C-I, D-III **2)** A-II, B-III, C-IV, D-I

3) A-III, B-II, C-IV, D-I **4)** A-IV, B-III, C-I, D-II

63. In chromyl chloride test for confirmation of Cl^- ion, a yellow solution is obtained. Acidification of the solution and addition of amyl alcohol and 10% H_2O_2 turns organic layer blue indicating formation of chromium pentoxide. The oxidation state of chromium in that is

1) +6 **2)** +5 **3)** +10 **4)** +3

64. The difference in energy between the actual structure and the lowest energy resonance structure for the given compound is

1) electromeric energy **2)** resonance energy

3) ionization energy **4)** hyperconjugation energy

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65. Given below are two statements :

Statement I : The electronegativity of group 14 elements from Si to Pb gradually decreases.

Statement II : Group 14 contains non-metallic, metallic, as well as metalloid elements.

In the light of the above statements, choose the most appropriate from the options given below :

1) Statement I is false but Statement II is true.

2) Statement I is true but Statement II is false.

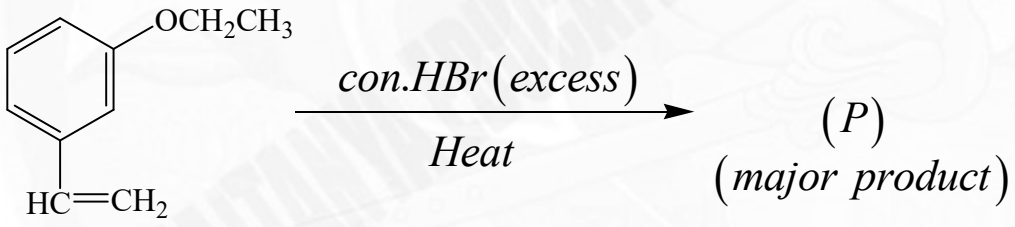
3) Both Statement I and Statement II are true.

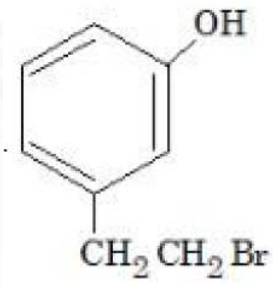
4) Both Statement I and Statement II are false.

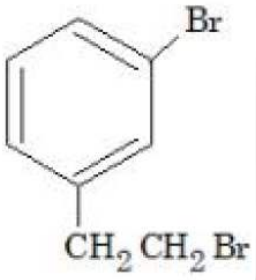
66. The correct set of four quantum numbers for the valence electron of rubidium atom (Z = 37) is:

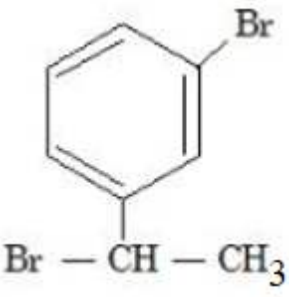
1) $5,0,0,+\frac{1}{2}$ **2)** $5,0,1,+\frac{1}{2}$ **3)** $5,1,0,+\frac{1}{2}$ **4)** $5,1,1,+\frac{1}{2}$

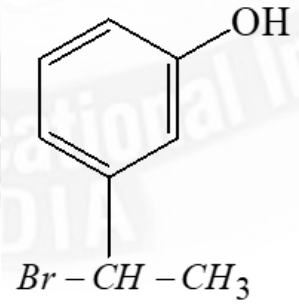
67. The major product(P) in the following reaction is



1) 

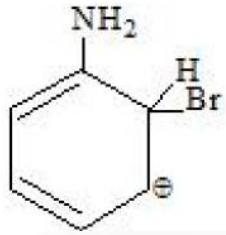
2) 

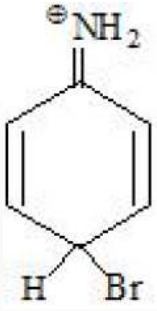
3) 

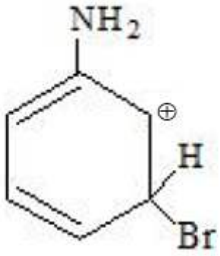
4) 

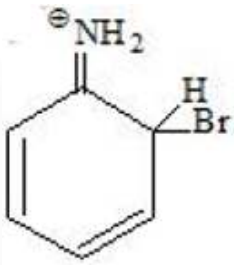
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68. The arenium ion which is not involved in the bromination of Aniline is

1) 

2) 

3) 

4) 

69. Appearance of blood red colour, on treatment of the sodium fusion extract of an organic compound with $FeSO_4$ in presence of concentrated H_2SO_4 indicates the presence of element/s

1) Br **2)** N **3)** N and S **4)** S

70. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Aryl halides cannot be prepared by replacement of hydroxyl group of phenol by halogen atom.

Reason R : Phenols react with halogen acids violently.

In the light of the above statements, choose the most appropriate from the options given below:

1) Both A and R are true but R is NOT the correct explanation of A

2) A is false but R is true

3) A is true but R is false

4) Both A and R are true and R is the correct explanation of A

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. For the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g), K_p = 0.492 atm \text{ at } 300K. K_c$ for the reaction at same temperature is___ $\times 10^{-2}$ (Given :$R = 0.082 Latmmol^{-1}K^{-1}$)

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72. A solution of H_2SO_4 is 31.4% H_2SO_4 by mass and has a density of 1.25g/mL. The molarity of the H_2SO_4 solution is ____ M (nearest integer) [Given molar mass of H_2SO_4=98gmol^{-1}]

73. Number of compounds with one lone pair of electrons on central atom amongst following is

$O_3, H_2O, SF_4, ClF_3, NH_3, BrF_5, XeF_4$

74. The mass of zinc produced by the electrolysis of zinc sulphate solution with a steady current of 0.015 A for 15 minutes is ____$\times 10^{-4}$g

(Atomic mass of zinc = 65.4 amu)

75. For a reaction taking place in three steps at same temperature, overall rate constant $K = \frac{K_1 K_2}{K_3}$ If Ea_1, Ea_2 and Ea_3 are 40, 50 and 60 kJ/mol respectively, the overall Ea is ____ kJ/mol.

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